

# Understanding the Chemical Bond in Semiconductor/MXene Composites: TiO<sub>2</sub> Clusters Anchored on the Ti<sub>2</sub>C MXene Surface

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First-principles calculations on titania clusters (TiO<sub>2</sub>)<sub>n</sub> (*n* = 5 and 10) supported on the pristine Ti<sub>2</sub>C (0001) surface were carried out to understand the properties of semiconductor/MXene composites with implications in (photo)-catalysis. The reported results reveal a high exothermic interaction accompanied by a substantial charge transfer with a concomitant, notorious, deformation of the titania nanoclusters. The analysis of the density of states analysis of the composite systems evidences a

metallic character with titania related states crossing the Fermi level. The picture of the chemical bonds is completed by the analysis of X-Ray Photoelectron Spectra (XPS) features, evidencing clear shifts of the C(1s) and O(1s) related peaks relative to the isolated systems that have a quite complex origin. This detailed analysis provides insights to experimentalists interested in the design and synthesis of these systems with possible applications in catalysis.

## 1. Introduction

Understanding the properties of semiconductor oxide and sulfide materials including TiO<sub>2</sub>,<sup>[1]</sup> WO<sub>3</sub>,<sup>[2]</sup> SrTiO<sub>3</sub>,<sup>[3]</sup> SnO<sub>2</sub>,<sup>[4]</sup> ZnO,<sup>[5]</sup> ZnS,<sup>[6]</sup> and Bi<sub>2</sub>S<sub>3</sub><sup>[7]</sup> has motivated the scientific community during the last decades because of the potential applicability of these materials in different photochemical and photocatalytic processes, with clear implications for energy and environment. Due to its excellent stability, high activity, chemical properties, and low cost,<sup>[8,9]</sup> titania (TiO<sub>2</sub>) has become the reference as photocatalyst. However, photoactive nanostructured titania polymorphs such anatase and rutile face important drawbacks limiting their applicability in photocatalysis.

The first handicap is directly related to the too large energy gap exceeding 3 eV,<sup>[1]</sup> which limits their ability to capture just 10% of sunlight photons in the ultraviolet (UV) region. The second one stands for the short lifetime of the photogenerated charge carriers (electrons and holes). The majority of these carriers are spontaneously deactivated reducing the yield of

certain photocatalytic reactions.<sup>[10,11]</sup> Despite these inherent limitations, titania still constitutes a common ingredient in the design of photoactive nanostructures, especially in combination with other materials to achieve practical photocatalytic applications such as water splitting,<sup>[12–14]</sup> fuel generation through CO<sub>2</sub> photoreduction,<sup>[15–20]</sup> hydrogen production from ethanol,<sup>[21]</sup> and photovoltaics solar cells,<sup>[22,23]</sup> just to name a few.

With the goal of improving the photocatalytic performance of titania derived materials, considerable efforts have been carried out in various directions. For instance, one can modify the morphology and dimension of TiO<sub>2</sub> nanoparticles,<sup>[1,24–26]</sup> doping with metal and non-metal atoms,<sup>[27–32]</sup> or coupling titania with metals or semiconductors creating complex structures such as composites and heterostructures.<sup>[33–39]</sup> This structural engineering procedure at the nanoscale makes it possible to modulate the energy gaps and to extend the lifetime of photogenerated carriers separating the photogenerated electrons and holes, and thus, increasing somehow its photocatalytic efficiency.<sup>[40]</sup> Among these strategies, the formation of double-phase materials has stood out promoting in principle an efficient separation between photogenerated charge carriers, thus increasing the possibility of carrying the redox reactions. Within this category, the formation of composites and heterostructures involving either a large phase of the host material with a scarce phase of another component or the junction of large phases of their components are among the most prominent strategies.<sup>[41]</sup> In this context, several studies have reported an enhancement of the titania photocatalytic properties by supporting it on carbonaceous nanomaterials,<sup>[42–44]</sup> by combining with another semiconductor,<sup>[7,19,38]</sup> by anchoring metal clusters such as Au,<sup>[21]</sup> by forming interfaces with perovskites,<sup>[35,36]</sup> polymeric materials like C<sub>3</sub>N<sub>4</sub>,<sup>[39]</sup> or even on another titania polymorph as in rutile-anatase composites.<sup>[34,45,46]</sup> Indeed, the commercial anatase-

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rutile nanostructure P90 composed of 92 and 8% of anatase and rutile, respectively, is extensively used in photocatalysis.<sup>[47]</sup>

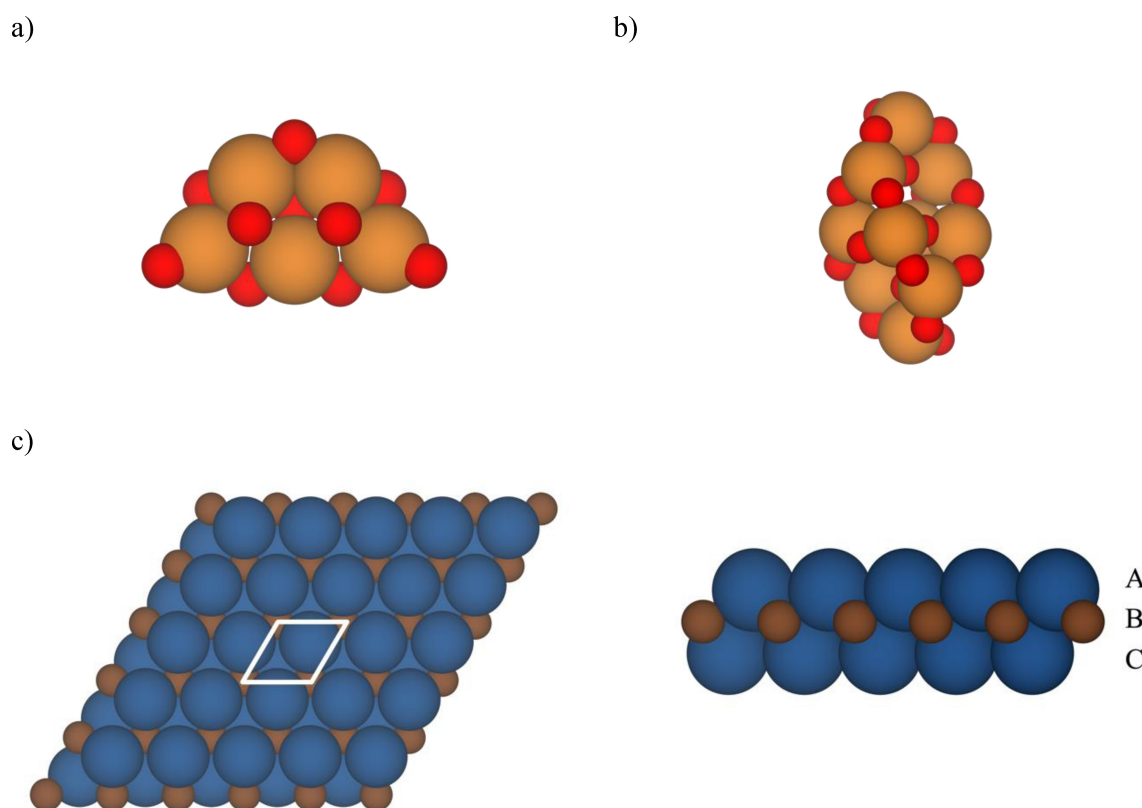
One of the promising new double-phase structures combines nanostructured  $\text{TiO}_2$  and MXenes, the latter being a relatively new family of two-dimensional (2D) materials involving metal carbides, nitrides, and/or carbonitrides,<sup>[48–50]</sup> usually displaying some terminating group. In fact, previous studies have introduced  $\text{TiO}_2$ /MXenes heterostructure as a type-II heterojunction that simultaneously facilitates the efficient separation of electrons and holes while preserving the redox ability of charge carriers.<sup>[51,52]</sup> This separation is rooted in the high electrical conductivity inherent in MXene materials.<sup>[53]</sup> Furthermore, the presence of a Schottky barrier in the heterojunction interfacial region limits the recombination of the photo-induced charge carriers and blocks the backward flow of electrons from the MXene to the  $\text{TiO}_2$ .<sup>[51,52,54,55]</sup> Altogether, these electronic properties derived from these novel structures put them in a privileged position for several applications with a large environmental and technological impact. In fact, there are several experimental works in the literature reporting applications of composite  $\text{TiO}_2$ /MXenes materials,<sup>[56–60]</sup> including photocatalytic  $\text{H}_2$  generation<sup>[61,62]</sup> and water splitting.<sup>[63,64]</sup> On the other hand, from a theoretical point of view, the information is really scarce. Here we highlight the work of Xu et al.<sup>[65]</sup> where the interaction between anatase and the functionalized  $\text{Ti}_2\text{C}$  and  $\text{Ti}_3\text{C}_2$  MXenes was investigated to shed light on the effect of the

MXene termination on the interfacial properties of the resulting heterostructure.

In the present work, we report a systematic study of the structural, electronic, and bonding properties of composites involving  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  nanoclusters supported on a pristine  $\text{Ti}_2\text{C}$  (0001) surface with the aim of investigating the chemical bond in these systems and the size effect on the composite which are thought to provide valuable information to experimentalists aiming at synthesizing and using these systems in a broad range of applications.

## 2. Materials models and computational details

With the focus on investigating the formation of  $\text{TiO}_2$ /MXene composites, the first task is to design suitable models. To this end, two different titania clusters with chemical formulas  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  were selected to study their interaction with the pristine  $\text{Ti}_2\text{C}$  (0001) surface of this MXene (Figure 1). This choice constitutes a representative case for titania clusters obtained from either bottom-up and top-down strategies.<sup>[26,66]</sup> Indeed, the structure selected for the  $(\text{TiO}_2)_5$  is a putative global minimum obtained by global optimization, while that of the  $(\text{TiO}_2)_{10}$  one is a bipyramid representative of anatase titania nanostructures often encountered in synthesized samples.<sup>[67]</sup> Note also that clusters of this size have been experimentally synthesized, and their spectroscopy properties determined.<sup>[68]</sup> On the other hand, the  $\text{Ti}_2\text{C}$  (0001) surface with ABC stacking is selected as it constitutes a well-known case.<sup>[69–71]</sup> In all cases, the clean MXene surface is considered. Note also that, even if



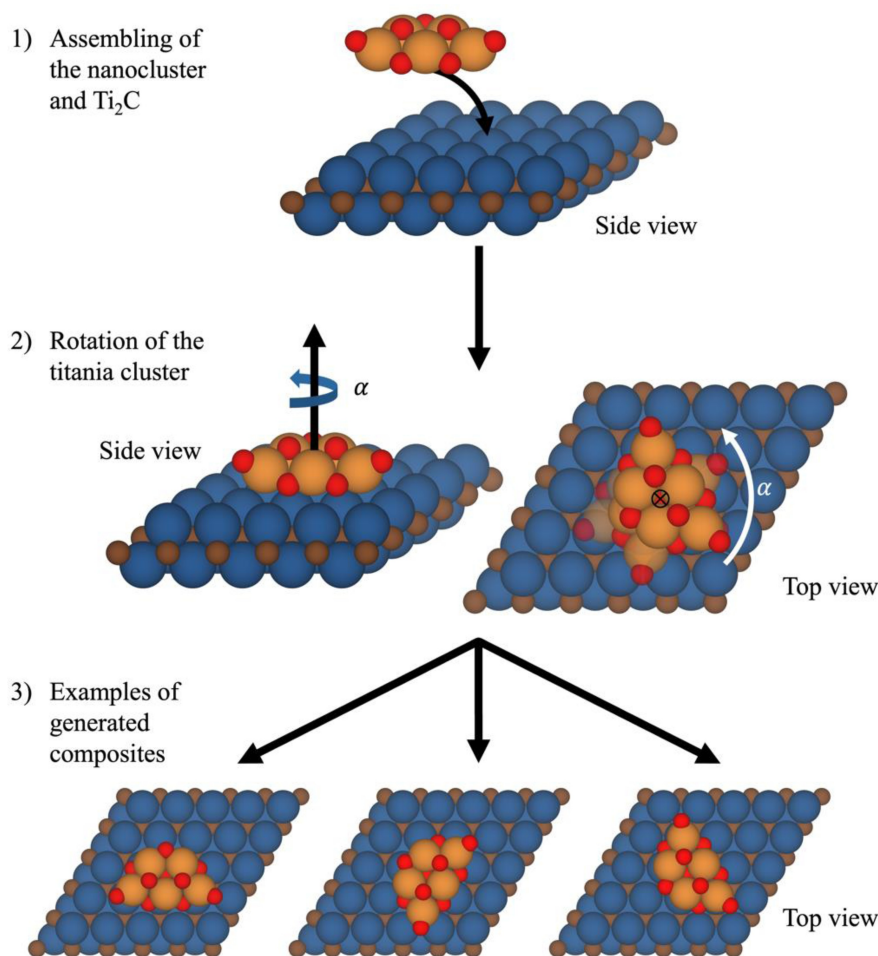
**Figure 1.** Atomic structures of planar  $(\text{TiO}_2)_5$  (top-left), bipyramidal  $(\text{TiO}_2)_{10}$  (top-right) nanoclusters, top (bottom-left) and side (bottom-right) views of the  $p(5 \times 5)$  supercell model of the  $\text{Ti}_2\text{C}$  (0001) surface where the white line shows the  $p(1 \times 1)$  cell. Note that the side view of  $\text{Ti}_2\text{C}$  exposes the ABC stacking. The orange and red spheres correspond to Ti and O atoms in  $\text{TiO}_2$  nanoclusters, while blue and brown spheres stand for Ti and C atoms of the  $\text{Ti}_2\text{C}$  MXene surface.

synthetic conditions leads to functionalized surfaces, it has been experimentally shown that clean surfaces can be prepared.<sup>[72,73]</sup>

To represent the  $\text{TiO}_2$  clusters supported on the  $\text{Ti}_2\text{C}(0001)$  surface, a slab model is used with supercells large enough to avoid any lateral interactions between the periodically repeated images. In particular,  $p(4\times 4)$ ,  $p(5\times 5)$ , and  $p(6\times 6)$  supercells were employed. By using the adsorption energies of  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  over each  $\text{Ti}_2\text{C}$  supercell at the same arbitrary configuration reported in Table S1 of the Supporting Information (SI) as a descriptor of the convergence with size, the  $p(5\times 5)$  supercell is considered appropriate to anchor the  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  clusters, since it provides a good balance in terms of computational cost and minimization of interactions between periodic replicas. This supercell is also suitable to anchor the  $(\text{TiO}_2)_{10}$  cluster since from  $(\text{TiO}_2)_5$  to  $(\text{TiO}_2)_{10}$ , the growth is in volume, not in base area, thus preserving a distance between replicas always larger than 6 Å. Test calculations for the  $(\text{TiO}_2)_{10}$  cluster using the larger  $p(6\times 6)$  supercell indicate that the adsorption energy exhibits a significant change in absolute terms of roughly 1 eV with respect to the  $E_{\text{ads}}$  arising from calculations with the  $p(5\times 5)$  supercell. However, in relative terms, the effect on  $E_{\text{ads}}$  is ~6%. For the purpose of the present work, aimed at a qualitative understanding, the choice of the  $p(5\times 5)$  supercell to study the interaction of the  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  clusters with the  $\text{Ti}_2\text{C}$  MXene surface represents a compromise between numerical accuracy and computational cost as, for the  $(\text{TiO}_2)_5$  cluster the

change in the  $E_{\text{ads}}$  values is negligible, while for the  $(\text{TiO}_2)_{10}$  is sufficiently small to reach sound conclusions.

To systematically explore the potential energy surface of  $(\text{TiO}_2)_n/\text{Ti}_2\text{C}$  systems, we develop an in-home Python code that allows us to compute the interaction energy between titania clusters and  $\text{Ti}_2\text{C}$  by using starting structures obtained by rotating the titania cluster over the  $\text{Ti}_2\text{C}(0001)$  surface.<sup>[74]</sup> Figure 2 depicts the workflow scheme used to generate the initial configurations of  $(\text{TiO}_2)_n/\text{Ti}_2\text{C}$  composites by varying the rotational angle. Briefly, the code first determines the center of mass (CoM) on the  $\text{Ti}_2\text{C}(0001)$  surface supercell and faces the  $(\text{TiO}_2)_n$  cluster and the  $\text{Ti}_2\text{C}$  surface. Next, the user selects an atom of the titania cluster to be anchored above the CoM. Finally, the code generates a set of different composites taking different rotational angles,  $\alpha$ , relative to an arbitrary configuration as the origin. Note that the range and step of degrees for the rotation of the cluster over the MXene (0001) surface are selected by the user to identify different adsorption sites and conformations, and every rotation is produced to the right, along the  $z$  axis, and relative to the arbitrary configuration chosen as origin. In the present study, we generate 14 different initial structures taking a rotating angle range from 0 to 140 sampling each  $10^\circ$ , and we labeled each structure with the initial rotation angle before the optimization step. Given the inherent hexagonal symmetry of this  $\text{Ti}_2\text{C}(0001)$  surface, this range guarantees a systematic, sufficient scan of the potential energy surface.



**Figure 2.** General scheme of the adsorption process of  $(\text{TiO}_2)_5$  cluster on  $\text{Ti}_2\text{C}(0001)$  surface done with our home-made Python code.<sup>[74]</sup> The program loads the cluster on top of  $\text{Ti}_2\text{C}$  center of mass (CoM). By rotating a certain degree,  $\alpha$ , different conformations are automatically generated. Color scheme is analogous to Figure 1.

Once the initial structures are generated, a structural relaxation is carried out in the framework of Density Functional Theory (DFT) using the well-known Perdew–Burke–Ernzerhof (PBE) functional,<sup>[75]</sup> which belongs to the family of Generalized Gradient Approximation (GGA) density functionals. PBE is a well-known and used DFT functional in material science and it provides a good description of closed-shell metallic systems,<sup>[76–9]</sup> thus, describes well the structural parameters of MXenes, as they are metallic systems. Note in passing that the contribution of dispersion to the interaction between the TiO<sub>2</sub> clusters and the Ti<sub>2</sub>C MXene surface, as estimated using the PBE–D3 approach, is one order of magnitude smaller than the PBE calculated adsorption energy. Consequently, the contribution of dispersion is not further considered as the focus of the present work is on the main trends rather than on reporting accurate values for systems which must be considered as appropriate models. In practice, the calculations are carried out using the Vienna *Ab Initio* Simulation Package (VASP).<sup>[80–82]</sup> The Kohn–Sham equations were numerically solved using a plane wave basis set used to expand the valence electron density with a kinetic energy cut-off of 415 eV. The Projector Augmented Wave (PAW) method,<sup>[83]</sup> as implemented by Kresse and Joubert,<sup>[84]</sup> was employed to account for the interaction between the core and valence electron densities. These PAW pseudopotentials include Ti 3p<sup>6</sup> 3d<sup>2</sup> 4s<sup>2</sup>, C 2s<sup>2</sup> 2p<sup>2</sup>, and O 2s<sup>2</sup> 2p<sup>4</sup> as valence electrons. Given the large size of the supercell, the numerical integrations in the reciprocal space have been carried using just the  $\Gamma$  point. In the present work, atomic relaxations were considered converged when the forces acting on the nuclei are all below 0.01 eV Å<sup>−1</sup> and a 10<sup>−5</sup> eV threshold was chosen as the electronic convergence criterion.

The thermodynamic stability of TiO<sub>2</sub>/Ti<sub>2</sub>C composites is analyzed using different the energetic descriptors defined in Eqs. 1–3,

$$E_{\text{ads}} = E_{\text{TiO}_2/\text{Ti}_2\text{C}} - E_{\text{Ti}_2\text{C}} - E_{\text{TiO}_2} \quad (1)$$

$$E^{\text{def}} = E_X^{\text{ads}} - E_X^{\text{free}} \quad (2)$$

$$E_{\text{adh}} = E_{\text{ads}} - E_{\text{Ti}_2\text{C}}^{\text{def}} - E_{\text{TiO}_2}^{\text{def}} \quad (3)$$

where  $E_{\text{ads}}$  stands for the adsorption energy;  $E_{\text{TiO}_2/\text{Ti}_2\text{C}}$  corresponds to the total energy of the composite;  $E_{\text{Ti}_2\text{C}}$  is the total energy of  $p(5 \times 5)$  Ti<sub>2</sub>C supercell, while  $E_{\text{TiO}_2}$  is the total energy of either (TiO<sub>2</sub>)<sub>5</sub> or (TiO<sub>2</sub>)<sub>10</sub>. Based on the definition of  $E_{\text{ads}}$ , the formation of the composites is favorable when  $E_{\text{ads}} < 0$  as it will correspond to an exothermic process. The second energetic criterion ( $E^{\text{def}}$ ) describes the energy required to deform the  $X$  building block, either the supported cluster ( $E_{\text{TiO}_2}^{\text{def}}$ ) or the MXene surface ( $E_{\text{Ti}_2\text{C}}^{\text{def}}$ ), from its optimum structure when isolated (free) to that on the composite system (ads). Finally, the adhesion energy ( $E_{\text{adh}}$ ) quantifies the strength of interaction between the already deformed fragments; clearly, the more negative  $E_{\text{adh}}$ , the stronger the interaction.

The nature of chemical bonding is extracted from different analyses. Firstly, we focus on net charges on the atoms or fragments as obtained from the Bader approach of Atoms-In Molecules (AIM) using the VASP-linked code provided by Henkelman *et al.*<sup>[85]</sup> This population analysis has been performed on each structure in its gas phase and composite conformations to rationalize the effect of the adsorption process on the bonding nature, quantifying the possible charge transfer between building blocks upon adsorption. Furthermore, the Charge Density Difference (CDD) and the Plane-Averaged CCD (PA-CCD) along the  $z$  direction have been computed following Eq. 4.

$$\Delta\rho = \rho_{\text{TiO}_2/\text{Ti}_2\text{C}} - \rho_{\text{Ti}_2\text{C}} - \rho_{\text{TiO}_2} \quad (4)$$

where  $\Delta\rho$  stands for the CDD;  $\rho_{\text{TiO}_2/\text{Ti}_2\text{C}}$  corresponds to the charge density of the TiO<sub>2</sub>/Ti<sub>2</sub>C composite; whereas  $\rho_{\text{Ti}_2\text{C}}$  and  $\rho_{\text{TiO}_2}$  correspond to the Ti<sub>2</sub>C and TiO<sub>2</sub> charge density, respectively. The CDD and PA-CDD complement the picture obtained from the Bader analysis providing an eye-catching view of the electron transfer between the components of the titania/MXene composites.

Finally, the picture of the chemical bond in these systems is completed by the analysis of the Core Level Binding Energies (CLBEs) of C(1s) and O(1s) before and after the formation of the composite. Here, we focus on the so-called initial state effects which do not take into account the electron density relaxation due to the presence of the generated core-hole and that, to a large extent, dominate the difference in CLBE of a given atom in different environments.<sup>[86–88]</sup> In the DFT framework, the Kohn–Sham orbital energies do not provide an approximation to the initial state absolute CLBE value as the Hartree–Fock orbital energies do.<sup>[89]</sup> However, it has been shown that these orbital energies provide reliable trends for CLBE shifts ( $\Delta\text{CLBEs}$ ).<sup>[90]</sup> In particular, it has been shown that the initial state effects govern the  $\Delta\text{CLBE}$  of bare MXenes.<sup>[91]</sup> Therefore, Kohn–Sham orbital energies can be used to provide estimates of initial state CLBEs with respect to a given reference. Accordingly, the CLBEs computed in the present work are always a measure of the initial state values which, in turn, are referred to the graphene C(1s). Here, the Kohn–Sham core level orbital energies of each system are as provided by the information from the corresponding PAW values at electronic convergence and are all referred to the vacuum level following Eq. 5.

$$\text{CLBE}_i = -[\varepsilon_i - V] \quad (5)$$

where  $\text{CLBE}_i$  corresponds to the CLBE of the  $i^{\text{th}}$ -level;  $\varepsilon_i$ , to the Kohn–Sham orbital energy for the corresponding core level  $i$ , and  $V$  is the electrostatic potential energy in the vacuum for each material. For a more complete the discussion regarding CLBEs, the work function ( $\phi$ ) of each solid material is also obtained following Eq. 6, and the Ionization Potential (IP) and Electron Affinity (AE) have been also estimated following Eqs. 7 and 8,

$$\phi = V - \varepsilon_F \quad (6)$$

$$\text{IP} = E(N-1) - E(N) \quad (7)$$

$$\text{EA} = E(N) - E(N+1) \quad (8)$$

where  $\varepsilon_F$  is the Fermi energy level for each material;<sup>[92,93]</sup>  $E(N-1)$ , the energy of the positively charged cluster with  $N-1$  electrons;  $E(N)$ , the energy of the neutral cluster with  $N$  electrons, and  $E(N+1)$ , the energy of the negatively charged cluster with  $N+1$  electrons.

To obtain a more accurate perspective of the electronic structure and overcome the limitations of GGA-derived functionals when describing electronic properties, *e.g.*, the well-known underestimation of the band gap,<sup>[94]</sup> we perform the electronic structure analysis from single-point calculations using the PBEx<sup>[95]</sup> hybrid functional carried out at the relaxed PBE titania/MXene systems; which is a usual strategy.<sup>[94,96]</sup> Note in passing that the PBEx functional includes a 12.5 % of non-local Fock exchange which is optimum for bulk titania polymorphs.<sup>[95]</sup> The use of PBEx allows one to reach a more accurate picture of the change in the electronic structure induced by the composite formation. This is achieved from the analysis of the density of states (DOS) of each composite and each building block computed by along with a denser grid of  $4 \times 4 \times 1$  special  $k$ -points.

### 3. Results and discussion

#### 3.1. Analysis of the formation of $(\text{TiO}_2)_n/\text{Ti}_2\text{C}$ composites

We start this section by commenting the energy related properties involved in the formation of  $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$  and  $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$  composites. The  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  clusters are placed above the  $p(5\times 5)$  supercell of the  $\text{Ti}_2\text{C}(0001)$  surface and the rotational angle along the  $z$  axis varied as explained in the previous section, followed by a full relaxation step. From the calculated values,  $E^{\text{def}}$  of each building block,  $E_{\text{ads}}$ , and  $E_{\text{adh}}$ , defined in Eqs. 1–3, are obtained and analyzed. Tables 1 and 2

compile these descriptors for  $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$  and  $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$  systems, respectively.

Starting with the  $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$  composite (Table 1), its formation is thermodynamically favored as it involves a highly exothermic  $E_{\text{ads}}$ . These highly negative  $E_{\text{ads}}$  are expected as is well-known across the literature that bare MXenes present a high reactive in the intrinsic (0001) surface, thus favoring adsorption of titania clusters. For the most stable configuration, here arbitrarily labeled as  $\alpha = 10^\circ$ ,  $E_{\text{ads}}$  is as large as  $-16.76$  eV. On the other hand, the lowest interaction between nanocluster and MXene corresponds to a rotational angle of  $90^\circ$  with  $E_{\text{ads}} = -13.02$  eV. Note that for  $\alpha = 40, 50$ , and  $60^\circ$ , the calculated  $E_{\text{ads}}$

**Table 1.** The adsorption energy ( $E_{\text{ads}}$ , in eV), the deformation energy of  $p(5\times 5)$   $\text{Ti}_2\text{C}$  and  $(\text{TiO}_2)_5$  ( $E_{\text{Ti}_2\text{C}}^{\text{def}}$  and  $E_{\text{TiO}_2}^{\text{def}}$ , respectively, in eV), and adhesion energy ( $E_{\text{adh}}$ , in eV), and net total charge of  $(\text{TiO}_2)_5$  titania nanocluster ( $Q_c$ , in  $e$ ) values upon adsorption for various configurations of  $p(5\times 5)$   $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$  composites with an  $\alpha$  degrees angle. These PBE energies are calculated following Eq. 1–3. The configuration with the largest  $E_{\text{ads}}$  is highlighted in bold.

| $\alpha$  | $E_{\text{ads}}$ | $E_{\text{Ti}_2\text{C}}^{\text{def}}$ | $E_{\text{TiO}_2}^{\text{def}}$ | $E_{\text{adh}}$ | $Q_c$        |
|-----------|------------------|----------------------------------------|---------------------------------|------------------|--------------|
| 0         | −15.56           | 0.85                                   | 17.24                           | −33.66           | −5.15        |
| <b>10</b> | <b>−16.76</b>    | <b>2.21</b>                            | <b>17.98</b>                    | <b>−36.95</b>    | <b>−6.02</b> |
| 20        | −15.79           | 1.78                                   | 16.55                           | −34.11           | −6.09        |
| 30        | −13.66           | 1.36                                   | 12.11                           | −27.14           | −4.69        |
| 50        | −16.60           | 0.67                                   | 10.34                           | −27.60           | −5.19        |
| 60        | −16.60           | 0.68                                   | 10.35                           | −27.63           | −5.19        |
| 70        | −14.85           | 1.17                                   | 11.64                           | −27.66           | −5.17        |
| 80        | −13.20           | 1.15                                   | 11.02                           | −25.36           | −4.49        |
| 90        | −13.02           | 1.30                                   | 13.00                           | −27.33           | −4.79        |
| 100       | −14.87           | 0.90                                   | 16.14                           | −31.91           | −5.72        |
| 110       | −14.87           | 0.91                                   | 16.13                           | −31.91           | −5.71        |
| 120       | −16.11           | 0.93                                   | 17.52                           | −34.56           | −5.81        |
| 130       | −14.86           | 0.91                                   | 16.10                           | −31.88           | −5.72        |
| 140       | −15.80           | 0.91                                   | 16.13                           | −32.84           | −5.89        |

**Table 2.** The adsorption energy ( $E_{\text{ads}}$ , in eV), the deformation energy of  $p(5\times 5)$   $\text{Ti}_2\text{C}$  and  $(\text{TiO}_2)_{10}$  ( $E_{\text{Ti}_2\text{C}}^{\text{def}}$  and  $E_{\text{TiO}_2}^{\text{def}}$ , respectively, in eV), and adhesion energy ( $E_{\text{adh}}$ , in eV), and net total charge of  $(\text{TiO}_2)_{10}$  titania nanocluster ( $Q_c$ , in  $e$ ) values upon adsorption for various configurations of  $p(5\times 5)$   $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$  composites with an  $\alpha$  degrees angle. These PBE energies are calculated following Eq. 1–3. The configuration with the largest  $E_{\text{ads}}$  is highlighted in bold.

| $\alpha$  | $E_{\text{ads}}$ | $E_{\text{Ti}_2\text{C}}^{\text{def}}$ | $E_{\text{TiO}_2}^{\text{def}}$ | $E_{\text{adh}}$ | $Q_c$        |
|-----------|------------------|----------------------------------------|---------------------------------|------------------|--------------|
| 0         | −17.21           | 0.96                                   | 21.69                           | −39.86           | −6.94        |
| 10        | −17.11           | 0.99                                   | 19.46                           | −37.57           | −6.76        |
| 20        | −17.11           | 0.99                                   | 19.49                           | −37.59           | −6.78        |
| <b>30</b> | <b>−19.83</b>    | <b>1.54</b>                            | <b>21.56</b>                    | <b>−42.92</b>    | <b>−7.71</b> |
| 40        | −15.00           | 1.26                                   | 11.13                           | −27.39           | −5.37        |
| 50        | −15.00           | 1.26                                   | 11.12                           | −27.38           | −5.38        |
| 60        | −13.54           | 1.10                                   | 7.99                            | −22.64           | −5.00        |
| 70        | −16.86           | 1.08                                   | 15.97                           | −33.91           | −6.67        |
| 80        | −14.74           | 0.63                                   | 17.04                           | −32.41           | −5.59        |
| 90        | −15.76           | 1.37                                   | 15.98                           | −33.11           | −6.07        |
| 100       | −15.76           | 1.47                                   | 15.22                           | −32.45           | −6.65        |
| 110       | −16.49           | 1.40                                   | 15.82                           | −33.72           | −6.68        |
| 120       | −16.98           | 2.41                                   | 21.24                           | −40.63           | −7.89        |
| 130       | −16.61           | 1.52                                   | 21.84                           | −39.97           | −6.95        |
| 140       | −16.76           | 0.98                                   | 14.35                           | −32.09           | −6.01        |

is  $-16.60$  eV implying that the separated optimization processes have converged to the same composite conformation. The same holds for the configurations at  $100$  and  $110^\circ$  with  $E_{ads} = -14.87$  eV. At this point, one must point out the arbitrary nature of the choice of a zero for the rotation angle  $\alpha$ . For instance, a more esthetical definition would consist of taking as zero the  $\alpha$  corresponding to the most stable configuration. However, this requires sampling the different structures corresponding to the possible rotations and just renaming the structures without adding new physical insight.

Next, we focus on  $E_{def}^{def}$  for each component of the system. The results in Table 1 clearly indicate that, upon adsorption on the  $Ti_2C$  (0001) surface, both systems deform but the titania cluster is the structural block that exhibits the largest deformation regardless of the rotational angle. Interestingly, the largest  $E_{TiO_2}^{def}$  (17.98 eV) corresponds to the configuration with the largest adsorption energy when  $\alpha = 10^\circ$ . The deformation of the  $Ti_2C$  (0001) is smaller but noticeable,  $E_{Ti_2C}^{def} = 2.21$  eV when  $\alpha = 10^\circ$ , and smaller for the rest of composite configurations, indicating an accommodation of each building block, especially of the cluster, to maximize the interaction. Regarding  $E_{adh}$ , the largest value ( $-36.95$  eV) corresponds to  $\alpha = 10^\circ$  while the lowest one ( $-25.36$  eV) is found for  $\alpha = 80^\circ$  (Table 1). All in all, this energy analysis shows that the adsorption site and relative orientation on the titania and MXene composite are important with the  $(TiO_2)_5/Ti_2C$  composite with largest adhesion and deformation energies being the most energetically favorable configuration.

For  $(TiO_2)_{10}/Ti_2C$  composite (Table 2), the aforementioned criterion holds for the most energetically favored configuration, revealing that the  $(TiO_2)_{10}/Ti_2C$  structure with  $\alpha = 30^\circ$  relative to our arbitrary origin is the most stable one with  $E_{ads} = -19.83$  eV. Conversely, the lowest interaction between the  $(TiO_2)_{10}$  cluster and the  $Ti_2C$  MXene ( $-13.54$  eV) corresponds to  $\alpha = 60^\circ$ . It is worth highlighting that for the initial configurations with  $\alpha = 40$  and  $50^\circ$  one finds the same  $E_{ads}$  value of  $-15.00$  eV, indicating that these rotation angles lead to the same composite configuration after relaxation. This is also the case for the configurations at  $\alpha = 10$  and  $20^\circ$  with  $E_{ads} = -17.11$  eV. Note, however, that despite the initial configurations with  $\alpha = 90$  and  $100^\circ$  present similar  $E_{ads}$  values, they do not possess the same geometry.

Regarding  $E_{def}^{def}$  and  $E_{adh}$  values for  $(TiO_2)_{10}/Ti_2C$  in Table 2, one readily finds that, again, the  $(TiO_2)_{10}$  cluster is the building block that experiments a larger deformation regardless of the initial configuration. This deformation allows the system to maximize the interaction. Note also that the  $Ti_2C$  (0001) surface is slightly affected upon adsorption. In addition, as in the previous case, the most energetically favored configuration exposes large  $E_{def}^{def}$  values of both cluster (21.56 eV) and  $Ti_2C$  (1.54 eV) and corresponds to  $\alpha = 30^\circ$ . Note, however, that the largest  $E_{Ti_2C}^{def}$  (2.41 eV) is found for  $\alpha = 120^\circ$  and the largest  $(TiO_2)_{10}$   $E_{TiO_2}^{def}$  (21.84 eV) for  $\alpha = 130^\circ$ . The results of this energy balance are that  $(TiO_2)_{10}/Ti_2C$  presents the largest interaction – i.e. the largest  $E_{adh}$  – at  $\alpha = 30^\circ$  ( $-42.92$  eV) while showing the lowest interaction strength with  $-22.64$  eV at  $60^\circ$ . To summarize, the  $(TiO_2)_{10}/Ti_2C$  structure at  $\alpha = 30^\circ$ , constitutes the most energeti-

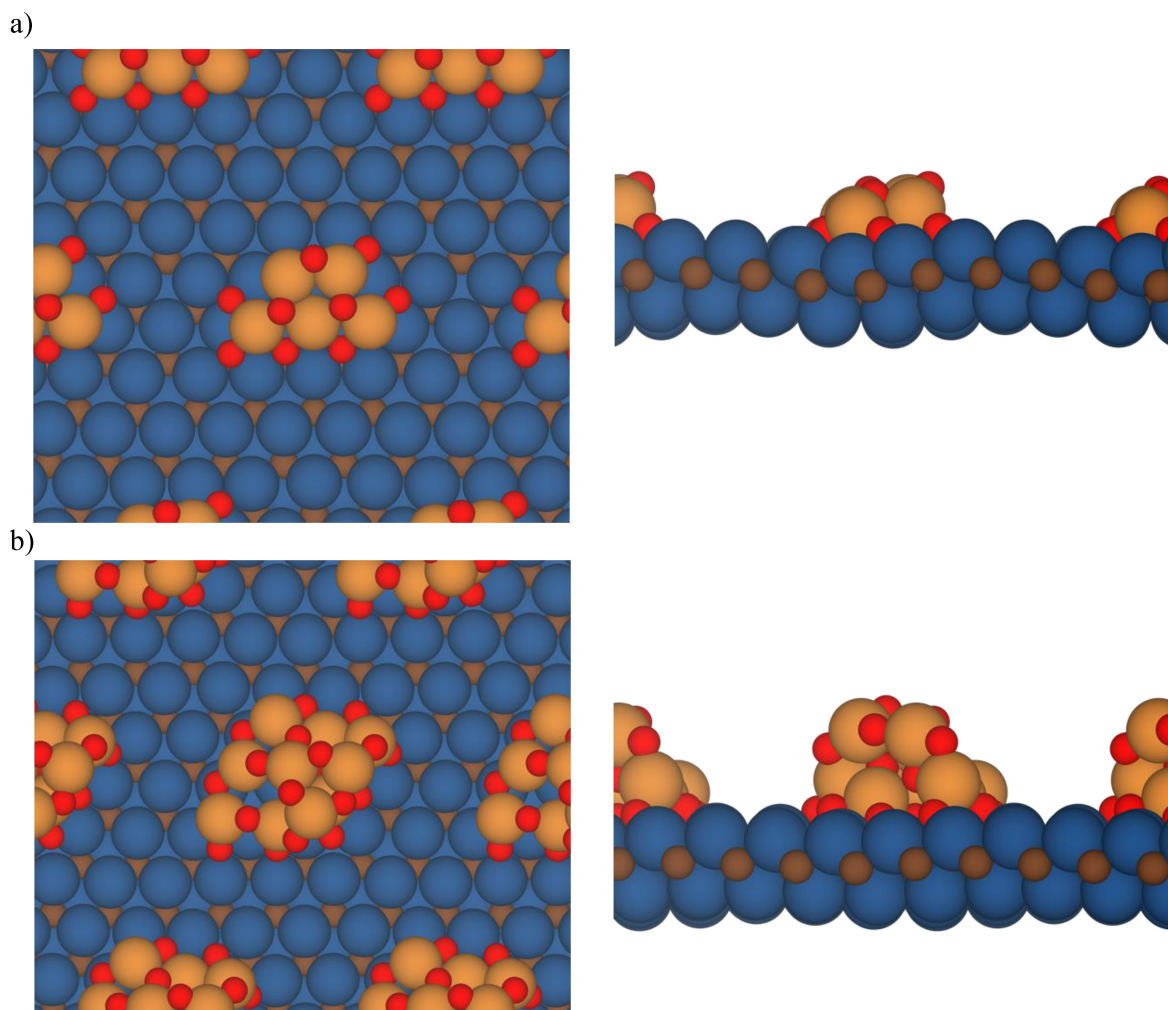
cally favorable configuration as presents the largest  $E_{ads}$  with the strongest interaction and one of the largest deformation energies. Here, we must point out again that the values of the rotational angle are arbitrary and used solely to define the different possible starting configurations, some of which become local minima and some of which converge to some of the other situations.

At this point, the most energetically favored configuration for both composites has been identified. Afterwards, we proceed to investigate the possible influence of the titania cluster size on the resulting  $E_{ads}$  and  $E_{adh}$ . To this end, the energies have to be normalized per atom in direct contact with the surface, the corresponding values are gathered in Table 3. The normalized adsorption energy ( $E_{ads/at}$ ) of the  $(TiO_2)_5$  cluster is 0.23 eV higher than the one of the  $(TiO_2)_{10}$  cluster, indicating that the  $(TiO_2)_5/Ti_2C$  formation is more exothermic than the  $(TiO_2)_{10}/Ti_2C$  despite the energy difference is not highly significant. The same pattern is followed by the normalized adhesion energy ( $E_{adh/at}$ ),  $(TiO_2)_5/Ti_2C$  presents a 0.56 eV higher  $E_{adh/at}$  than  $(TiO_2)_{10}/Ti_2C$  one, thus, the  $(TiO_2)_5$  cluster has a stronger interaction with the surface than the  $(TiO_2)_{10}$  one. The larger values for the  $(TiO_2)_5$  cluster are likely to be due its smaller size with a larger number of low coordinated atoms. At this point, one could wonder whether planar structures are possible for the supported  $(TiO_2)_{10}$  clusters. To this end, several planar structures were designed, supported on the MXene surface and optimized. However, all attempts to find a stable, planar  $(TiO_2)_{10}$  supported on  $Ti_2C$  failed meaning that, at least in this regime, three-dimensional supported clusters are more stable than the two-dimensional ones.

The fully relaxed geometries of the most energetically favorable configurations of each composite before and after adsorption are presented in Figures 1 and 3, respectively. These clearly show that the interaction between the  $(TiO_2)_5$  and  $(TiO_2)_{10}$  clusters and the  $Ti_2C$  MXene leads to significant changes in the morphology of both cluster and  $Ti_2C$  in concordance with the deformation values from Tables 1 and 2. Indeed, both titania clusters lose their initial geometry, and in fact, in the case of  $(TiO_2)_5$  cluster, the adsorbed geometry of the cluster seems to follow the MXene ABC stacking. Regarding the changes in the morphology upon adsorption for  $(TiO_2)_5/Ti_2C$  composite, the average distances exhibit noticeable changes. Thus, some Ti–Ti and Ti–O distances inside the  $(TiO_2)_5$  cluster increase by  $\sim 0.4$  Å while others are not changed at all. Interestingly, upon adsorption, the O atoms occupy two high-symmetry sites on the  $Ti_2C$  surface, hollow metal and hollow

**Table 3.** The number of atoms from each cluster in direct contact with the  $Ti_2C$  surface ( $N$ ) at  $p(5 \times 5)$   $(TiO_2)_5/Ti_2C$  and  $p(5 \times 5)$   $(TiO_2)_{10}/Ti_2C$  composites, and the adsorption and adhesion energies of the most energetically favorable configuration of each composite normalized by  $N$  ( $E_{ads/at}$  and  $E_{adh/at}$ , respectively, in eV/atom in direct contact with the surface). These energies are calculated from values in Tables 1 and 2.

| System               | $N$ | $E_{ads/at}$ | $E_{adh/at}$ |
|----------------------|-----|--------------|--------------|
| $(TiO_2)_5/Ti_2C$    | 12  | $-1.40$      | $-3.08$      |
| $(TiO_2)_{10}/Ti_2C$ | 17  | $-1.17$      | $-2.52$      |



**Figure 3.** Top (left) and side (right) view of fully relaxed structures of (a)  $p(5 \times 5)$   $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$  with  $\alpha = 10^\circ$ , and (b)  $p(5 \times 5)$   $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$  with  $\alpha = 30^\circ$ . Color scheme is analogous to Figure 1.

carbon sites, so that two O atoms change plane to maximize the contact with the  $\text{Ti}_2\text{C}(0001)$  surface. Regarding structural changes in the  $\text{Ti}_2\text{C}$  substrate produced by the interaction with the  $(\text{TiO}_2)_5$  cluster, the Ti–Ti distances between Ti atoms on the same layer located at the interface decrease by  $\sim 0.2 \text{ \AA}$  whereas those far from the interphase increase by  $\sim 0.2 \text{ \AA}$ . The distance between Ti atoms from different layers and Ti–C distances remain intact. In the case of the  $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$  composite, the Ti–Ti and Ti–O distances of the  $(\text{TiO}_2)_{10}$  unit increase upon adsorption by  $\sim 0.3$  and  $\sim 0.2 \text{ \AA}$ , respectively. Interestingly, this increase in the interatomic distances in the  $(\text{TiO}_2)_{10}$  cluster is smaller than for the  $(\text{TiO}_2)_5$  one. This is because, in the larger  $(\text{TiO}_2)_{10}$  cluster, the larger stability implies higher difficulty in deforming it when put in contact with the  $\text{Ti}_2\text{C}(0001)$  surface. Moreover, this is in agreement with the deformation energies discussed above which are even larger for the larger cluster and smaller per  $\text{TiO}_2$  unit. In the  $(\text{TiO}_2)_{10}$  case, all O atoms also occupy high-symmetry sites of the MXene surface, nearly all occupy hollow metal sites with a few exceptions, some of them are at hollow carbon and bridge sites. Again, in the  $\text{Ti}_2\text{C}$  building block for this composite, only distances between Ti

atoms from the same layer at the interface are changed roughly by  $\sim 0.2 \text{ \AA}$ , whereas Ti–Ti from different layers and Ti–C distances remain unchanged as in the  $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$  composite.

### 3.2. The nature of the chemical bond between $(\text{TiO}_2)_n$ clusters and the $\text{Ti}_2\text{C}$ surface

The Bader charge analysis is carried out to identify changes in the chemical bond when the composite is formed. This population analysis is performed on every structure before and after the composite formation which allows one to quantify the charge transfer and its transferring direction. Tables 1 and 2 contain the cluster Bader charges for each composite and adsorption site. Since the titania clusters in gas phase are neutral, the negative values of the accumulated Bader charges in the supported titania nanocluster indicates that there is a systematic charge transfer from the MXene to titania nanoclusters regardless of the adsorption site for both titania nanoclusters (see Tables 1 and 2). Focusing firstly on the  $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$  composite, one clearly see that the adsorption site and

configuration has a strong influence on the total charge transfer. For instance, the largest charge transfer occurs at  $\alpha = 10^\circ$ , the most favorable configuration, and  $\alpha = 20^\circ$  with  $-6.02$  and  $-6.09$  e, respectively, whereas the smallest one corresponds to at  $\alpha = 30$  and  $\alpha = 80^\circ$ , with  $-4.69$  and  $-4.49$  e accumulated on the  $(\text{TiO}_2)_5$  cluster, respectively.

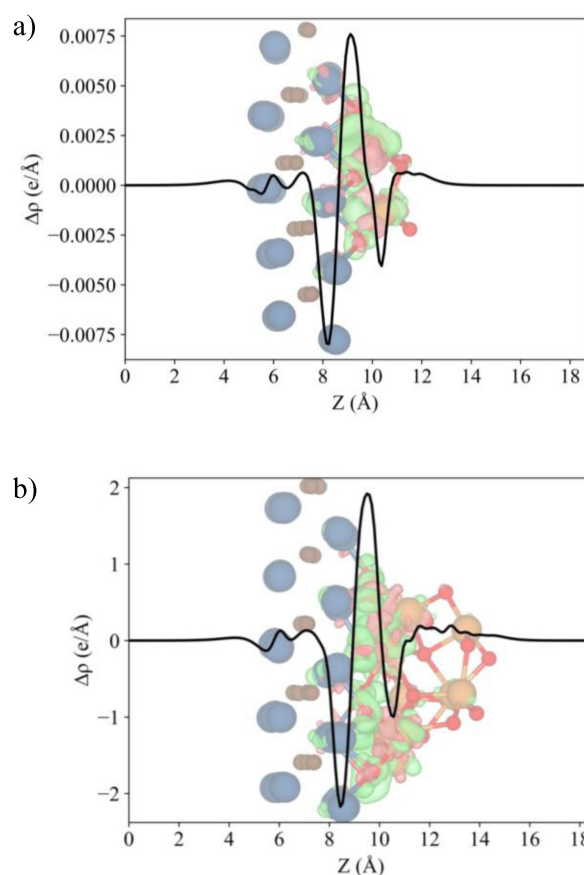
A more accurate and detailed analysis of the charge transfer in this system can be made by focusing on the Bader charges for each individual atom available in Table S2 from SI. The Ti atoms of the isolated  $\text{Ti}_2\text{C}$  MXene and isolated titania cluster have average Bader charges of  $1.17$  and  $2.10$  e Ti per atom, respectively, whereas for C and O, one gets  $-2.34$  and  $-1.05$  e per atom, respectively. However, upon adsorption, the nanocluster Ti atom charges decrease from  $2.10$  e to an average of  $\sim 1.40$  e per atom, consistent with a reduction of Ti atoms belonging to titania nanocluster and adopting nearly the same charge as those Ti atoms in  $\text{Ti}_2\text{C}$  MXene. It is worth mentioning that the O charges are only slightly affected by the adsorption process by decreasing an average  $\sim 0.30$  e per O atom in direct contact with the surface, indicating that Ti atoms of the nanocluster are the main ones accepting the charge transfer. This charge transfer to the supported cluster implies the subsequent oxidation of Ti atoms of the  $\text{Ti}_2\text{C}$  MXene observed, although the change with respect to the pristine surface is small due to the large number of Ti atoms in the  $\text{Ti}_2\text{C}$  supercell. The interaction between the cluster and the MXene also implies a reorganization of the charges of the latter with the C atoms closer to the interface – just below the cluster – are less negatively charged ( $-2.18$  e) than on the isolated MXene case ( $-2.34$  e), thus becoming slightly oxidized with respect to rest of C atoms.

For the  $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$  system one finds the same trend as the  $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$  one for the charge transfer. The influence of adsorption sites also leads to variations in charge transfer values for each configuration. For example, as shown in Table 2, the largest charge transfers are observed at  $\alpha = 30^\circ$ , the most favorable configuration, and  $\alpha = 120^\circ$ , with values of  $-7.71$  and  $-7.89$  e, respectively. In contrast, the smallest charge transfers occur at  $\alpha = 40$  and  $\alpha = 60^\circ$ , with values of  $-5.37$  and  $-5.00$  e, respectively.

The broken-down Bader charges for each atom from  $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$ , available in Table S3 from SI, also follow the pattern discussed for the  $(\text{TiO}_2)_5$  nanocluster. The Ti atoms of the  $\text{Ti}_2\text{C}$  MXene are slightly oxidized, and the charge is transferred towards the Ti atoms of the  $(\text{TiO}_2)_{10}$  nanocluster, in such a way that all Ti atoms adopt nearly the same charge. However, due to the larger number of atoms of this cluster, not all  $(\text{TiO}_2)_{10}$  cluster Ti atoms are equally affected, the reduction involves essentially the Ti atoms of the nanocluster that are in direct contact with the  $\text{Ti}_2\text{C}$  (0001) surface, by decreasing their positive charge by an average  $0.63$  e per atom. This finding reveals a localized charge rearrangement near the interface. The reduction of the rest Ti atoms in titania nanocluster is negligible because they are located far from the interface. In addition, the O atoms are also slightly reduced, as in  $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$ , by decreasing their average charge by  $\sim 0.29$  e. The C atoms in  $\text{Ti}_2\text{C}$  are also affected by the presence of the  $(\text{TiO}_2)_{10}$  nanocluster

over the MXene surface in the same way that occurs for the smaller cluster, in this case, the C atoms near the interface are slightly oxidized by increasing their averaged charge per C atom below the cluster by  $\sim 0.17$  e per C atom. At this point one may be tempted to relate the calculated Bader charges to the coordination number of the involved atoms. Nevertheless, the large number of undercoordinated atoms in the supported clusters makes it difficult to relate the coordination number, which exhibits significant variations, to the Bader charge where the variations are less pronounced.

The Bader analysis is complemented by the CDD and PA-CDD along the  $z$  direction available in Figures 4a,b for both composites. This visual analysis support the conclusions obtained from the Bader analysis, a charge rearrangement occurs around the interface of the systems: a depletion of charge density is observed at the MXene Ti atoms and an accumulation, at the Ti atoms from the titania clusters. Note that due to the larger number of atoms of the titania  $(\text{TiO}_2)_{10}$  nanocluster in direct contact with the surface (Table 3), the charge transfer is larger in absolute value than the one that occurs at the  $(\text{TiO}_2)_5$  nanocluster. Note in passing that the charge transfer induced by the interface formation from the  $\text{Ti}_2\text{C}$  MXene to the titania clusters could generate a build-in



**Figure 4.** Plane-averaged differential charge densities ( $\Delta\rho$  in  $\text{eV}/\text{\AA}$ ) along the  $z$  direction for (a)  $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$   $\alpha = 10^\circ$ , and (b)  $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$   $\alpha = 30^\circ$ . Charge density differences with an isosurface value of  $0.01 \text{ e}/\text{\AA}^3$  are depicted inset where green and red isosurfaces corresponding to accumulation and depletion charges, respectively. Color code is as in Figure 1.

electric field near the interface that can facilitate a possible photogenerated charge carrier separation.<sup>[97]</sup>

### 3.3. Electronic structure analysis

To circumvent the drawbacks of GGA functionals in the description of the electronic structure of insulating material, results in this subsection are obtained using the hybrid PBEx functional with the corresponding calculations carried out at the PBE structures, a good compromise between accuracy and computational requirements. The projected DOS per atomic orbital of each building block and composites are gathered in Figure 5. These plots allow one to unveil how the electronic structure of each component is affected by the interaction. First, focusing on the titania nanoclusters, both show a quite insulating DOS, presenting an energy gap of 3.8 eV for (TiO<sub>2</sub>)<sub>5</sub> and 3.5 eV for (TiO<sub>2</sub>)<sub>10</sub> (see Figures 5a and 5b), also displaying discrete states typical of finite systems. The valence states mainly belong to the O(2p), and the unoccupied ones, to Ti(3d). These available empty Ti(3d) states could justify the charge transfer largely directed towards the TiO<sub>2</sub> Ti atoms. In addition, some unoccupied states also have O(2p) character, thus justifying the slight reduction of the supported clusters O atoms upon adsorption. Note also that the energy gaps are slightly larger than the anatase bulk structure 3.2 eV,<sup>[98]</sup> due to the quantum confinement. On the other hand, the electronic structure representation of the Ti<sub>2</sub>C MXene shows a gapless electronic structure, indicating a metallic character with broad bands occupied band dominated by the contribution of Ti(3d) near the Fermi level (*cf.* Figure 5c) as in agreement with previous studies.<sup>[71,74]</sup>

Upon adsorption, the total DOS of both composited exhibits an overall metallic character (Figures 5d and 5e) with the O(2p) states lying lower with respect to the Fermi level, while the Ti(3d) ones, are around it. In fact, the main contribution to the conduction band states remains dominated by Ti(3d) derived levels. Projecting the total DOS onto each building block, one can observe that the main contribution to the band crossing the Fermi level comes from the metallic Ti<sub>2</sub>C component (Figures 5f and 5g), nonetheless, a wide Ti(3d) band from the titania nanoclusters is also clearly seen, indicating that the titania cluster and the Ti<sub>2</sub>C electronic structure significantly merge in such a way that the composite system exhibit a

metallic character. Finally, note that for the larger, supported, (TiO<sub>2</sub>)<sub>10</sub> cluster, the number of Ti(3d) band above the Fermi level is larger than for the (TiO<sub>2</sub>)<sub>5</sub> case which is simply a consequence of the larger number of Ti atoms in the (TiO<sub>2</sub>)<sub>10</sub> nanocluster.

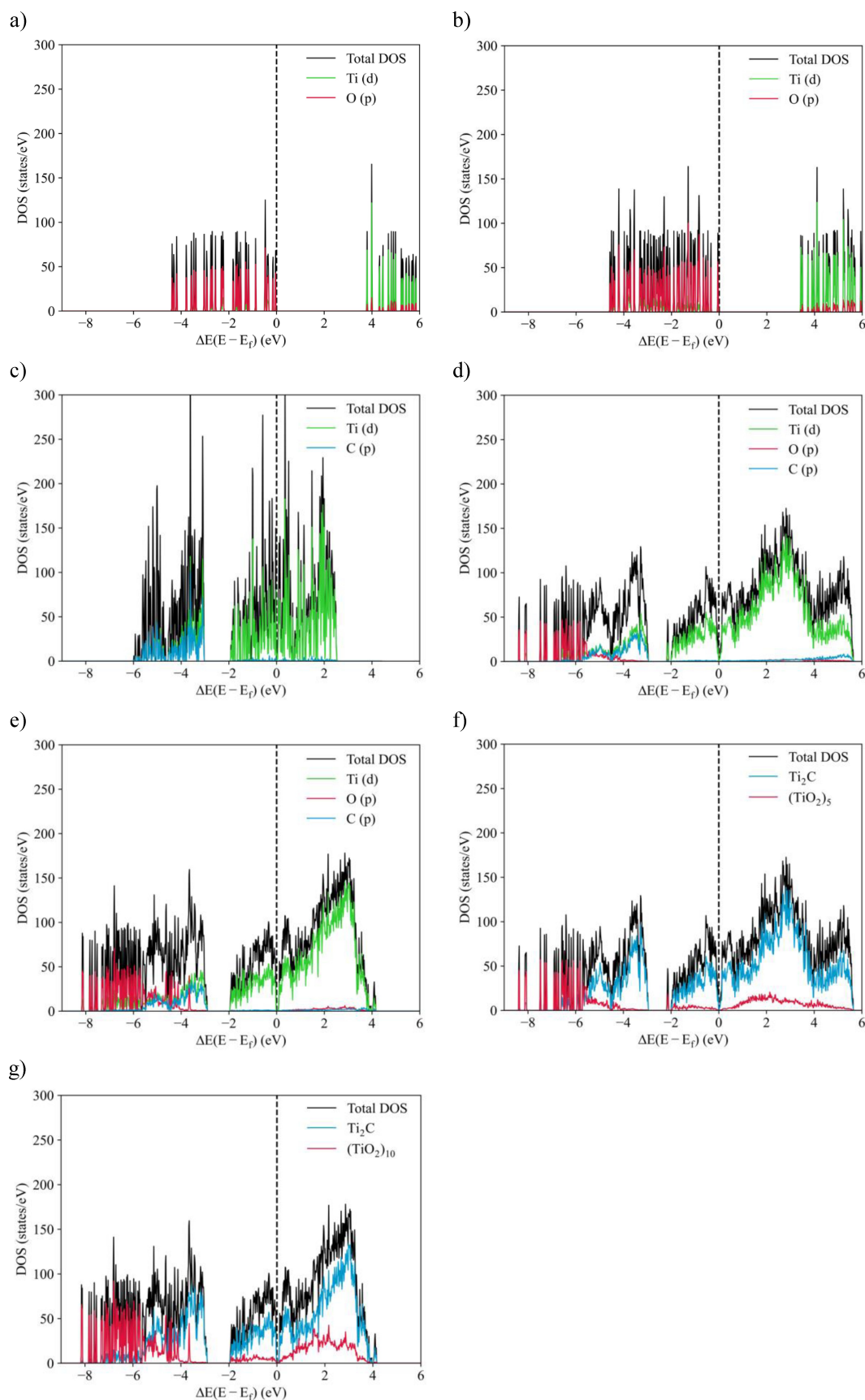
### 3.4. Analysis of the core level binding energies

The C(1s) and O(1s) CLBEs XPS characteristic peaks have been simulated for both composites at the most energetically favored configuration using the initial state approximation to analyze the shifts resulting from the formation of these composites, which can be of interest for experimental characterization of these composite. Table 4 compiles the CLBE initial states and  $\Delta$ CLBE values derived from the difference between the C(1s) and O(1s) CLBE levels with respect to graphene C(1s), all referenced to the vacuum level of each system using Eq. 5. Note, however, that the thus calculated CLBEs cannot be trusted,<sup>[90]</sup> and are included here just for comparison. In fact, the initial state CLBE for C(1s) in graphene, estimated from Kohn–Sham orbital energies and referenced to the vacuum level, is 270.5 eV which value is significantly lower than the experimental one for graphene (284.6 eV),<sup>[99]</sup> in spite of neglecting the energy relaxation in response the presence of the core hole; for a more complete discussion, the reader is referred to previous work.<sup>[90]</sup> In addition, despite the C(1s) CLBE of the bare Ti<sub>2</sub>C is not available, the experimental value of the functionalized Ti<sub>2</sub>CT<sub>x</sub> C(1s) CLBE is reported to be 281.9 eV,<sup>[100]</sup> which is larger than the hereby theoretically predicted and referred to its vacuum level value of 269.3 eV, which is in agreement with previous theoretical studies.<sup>[91]</sup> Importantly, the error for both graphene and Ti<sub>2</sub>C C(1s) CLBE is similar and systematic, making the calculated initial state  $\Delta$ CLBEs relatively reasonable.

The  $\Delta$ CLBE C(1s) values for the MXene building block from Table 4 show a non-negligible influence of the cluster adsorption on the C(1s) with a shift of −0.2 and −0.3 eV respect the isolated Ti<sub>2</sub>C C(1s) CLBE when (TiO<sub>2</sub>)<sub>5</sub> and (TiO<sub>2</sub>)<sub>10</sub> clusters are adsorbed on the surface, respectively. Note that, as discussed in the previous section, the C atoms of Ti<sub>2</sub>C become slightly oxidized when supporting the (TiO<sub>2</sub>)<sub>5</sub> and (TiO<sub>2</sub>)<sub>10</sub> clusters which, according solely to charge transfer arguments, this should lead to higher, not lower, CLBEs. To shed light on this apparent contradiction, the C(1s) CLBE has also been calculated for the

**Table 4.** Initial state C(1s) and O(1s) CLBE relative to the vacuum energy level (see Eq. 5),  $\Delta$ CLBEs relative to C(1s) CLBE of graphene for each system computed with PBE for (TiO<sub>2</sub>)<sub>5</sub>, (TiO<sub>2</sub>)<sub>10</sub>, Ti<sub>2</sub>C, and the respective composites at the most energetically stable configuration. The CLBEs are selected from atoms close to the surface. In addition, the work function ( $\phi$ ) is calculated for each solid system, and for each cluster system, the Ionization Potential (IP) and Electron Affinity (EA) have been estimated following Eqs. 6–8. The units of all energetic parameters are in eV.

| Structure                                            | C(1s) CLBE | C(1s) $\Delta$ CLBE | O(1s) CLBE | O(1s) $\Delta$ CLBE | $\phi$ | IP  | EA  |
|------------------------------------------------------|------------|---------------------|------------|---------------------|--------|-----|-----|
| (TiO <sub>2</sub> ) <sub>5</sub> , gas phase         | –          | –                   | 511.0      | 240.5               | –      | 8.1 | 3.1 |
| (TiO <sub>2</sub> ) <sub>10</sub> , gas phase        | –          | –                   | 511.0      | 240.5               | –      | 7.0 | 3.0 |
| Ti <sub>2</sub> C                                    | 269.3      | −1.2                | –          | –                   | 4.3    | –   | –   |
| (TiO <sub>2</sub> ) <sub>5</sub> /Ti <sub>2</sub> C  | 269.1      | −1.4                | 512.3      | 241.7               | 4.1    | –   | –   |
| (TiO <sub>2</sub> ) <sub>10</sub> /Ti <sub>2</sub> C | 269.0      | −1.5                | 511.6      | 241.1               | 4.1    | –   | –   |



**Figure 5.** Projected DOS per atomic orbital (in states/eV) for (a)  $(\text{TiO}_2)_5$ , (b)  $(\text{TiO}_2)_{10}$ , (c)  $p(5 \times 5)$   $\text{Ti}_2\text{C}$ , (d)  $p(5 \times 5)$   $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$ , (e)  $p(5 \times 5)$   $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$  and projected DOS per building block (in states/eV) for (f)  $p(5 \times 5)$   $(\text{TiO}_2)_5/\text{Ti}_2\text{C}$ , and (g)  $p(5 \times 5)$   $(\text{TiO}_2)_{10}/\text{Ti}_2\text{C}$ . All DOS are obtained from single-point calculations with the PBE functional over the PBE structures. The energies are referenced to the Fermi level (dashed line in eV).

isolated MXene at the adsorbed geometry. At the isolated  $(\text{TiO}_2)_5$ - and  $(\text{TiO}_2)_{10}$ -adsorbed  $\text{Ti}_2\text{C}$  geometries, the  $\text{C}(1s)$  CLBEs are at 269.4 eV for both geometries which almost coincides with the value for isolated  $\text{Ti}_2\text{C}$ , indicating that the structural change induced by the presence of the adsorbed cluster has a negligible effect on the  $\text{C}(1s)$  CLBE. Note that, in any case, the predicted shifts are very small, almost in the limit of detection even with a synchrotron radiations source, indicating that the  $\text{C}(1s)$  is not significantly affected by the presence of the clusters at the surface. This also show that XPS focusing on the  $\text{C}(1s)$  of these composite systems will hardly provide useful information.

On the other hand, the effect of cluster adsorptions on  $\text{O}(1s)$  is more noticeable. Upon adsorption, the shifts with respect to the isolated nanoclusters are +1.2 and +0.6 eV for the  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  clusters, respectively. These shifts of the  $\text{O}(1s)$  CLBEs to a higher binding energy. The origin of these shifts is unclear as the charge transfer is from  $\text{Ti}_2\text{C}$  to each nanocluster meaning that one would expect a shift to lower energies. For the isolated cluster at the adsorbed geometry the calculated  $\text{O}(1s)$  CLBEs are of 512.3 and 510.3 eV, respectively, sufficiently different from the 511.0 eV value for the isolated clusters at their most stable structure. These results show a delicate interplay between structure and charge transfer on defining the CLBEs and their corresponding shifts, warning also about oversimplistic interpretations in these complex systems. Here it is worth insisting on the fact that the predicted XPS shifts upon adsorption are not in the direction that one would expect from the calculated charge transfer meaning that, in this case, charge transfer is not the origin of the shift.

Finally, we comment on the  $\phi$  of the  $\text{Ti}_2\text{C}$  MXene and  $\text{TiO}_2/\text{Ti}_2\text{C}$  composite systems, and also the IP and EA of the isolated nanoclusters in their most stable structure (*cf.* Eqs. 6–8). These quantities provide information about the energy cost to extract (or add) one electron from the system of interest. Upon composite formation, the vacuum levels align, causing the building block with the lower  $\phi$  to transfer electrons to the one with the higher IP or *vice versa*. One can notice that the  $\text{Ti}_2\text{C}$   $\phi$  (4.3 eV) is lower than the IPs for the  $\text{TiO}_2$  nanoclusters (8.1 and 7.0 eV for the  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  clusters, respectively). Therefore, one may expect that the direction of electron transfer will be from  $\text{Ti}_2\text{C}$  to the titania nanoclusters as found in the calculations. It is worth mentioning that the EA for each cluster is noticeably lower than the IPs, meaning that the clusters are more likely to accept electrons from  $\text{Ti}_2\text{C}$  rather than donate them. Altogether, this aligns with the findings from chemical and electronic analysis from sections 3.2 and 3.3, where the charge transfer from the MXene to both nanoclusters is rationalized. Finally, note that the  $\phi$  of both composites remains nearly unchanged at 4.1 eV, indicating no significant alteration in the  $\text{Ti}_2\text{C}$   $\phi$  upon adsorption as expected.

## 4. Conclusions

The structure, bonding nature, and electronic properties of  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  clusters, supported on the bare  $\text{Ti}_2\text{C}$  (0001)

surface is described in terms of DFT-based calculations. For the most stable adsorbed structures, the interaction is highly exothermic energy with adsorption energy values of −16.76 eV and −19.83 eV for  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$ , respectively. To a large extent, the difference between the interaction energy of the cluster is due to the difference in the number of cluster atoms in contact with the  $\text{Ti}_2\text{C}$  surface. This large interaction is interpreted in terms of cost-benefit based on deformation and adhesion energies. In general, the largest interaction is also accompanied by a larger deformation of the titania nanoclusters.

The analysis of the chemical bond between the supported titania clusters and the  $\text{Ti}_2\text{C}$  MXene reveals a large charge transfer from  $\text{Ti}_2\text{C}$  towards the  $\text{TiO}_2$  nanoclusters of −6.02 and −7.71 e for  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$ , respectively, affecting mostly the titania Ti atoms in direct contact with the  $\text{Ti}_2\text{C}$  surface which are reduced, accepting ~1.40 and ~0.63 e per Ti atom for  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$ , respectively. Consequently, the charge reorganization involves essentially the interface region. It is worth mentioning that the O atoms are also slightly reduced with a concomitant subsequent oxidation of  $\text{Ti}_2\text{C}$  Ti atoms in contact with the supported cluster, thus preserving the charge neutrality of the formed composite. Similar conclusions are achieved from the analysis of the CDD and PA-CDD, all showing a strong charge rearrangement at the interface, *i.e.* a depletion of charge density is observed at the Mxene Ti atoms and an accumulation, at the Ti atoms from the titania nanoclusters.

The electronic structure of each composite was studied employing the PBE hybrid density functional. The analysis of the DOS shows unoccupied  $\text{Ti}(3d)$  states of titania nanoclusters, in accordance with the observed charge transfer direction. It is worth highlighting the small amount of unoccupied  $\text{O}(2p)$  states that also leads to a small reduction of the O atoms upon adsorption. For isolated  $(\text{TiO}_2)_5$  and  $(\text{TiO}_2)_{10}$  nanoclusters systems, the predicted energy gaps are 3.8 and 3.5 eV, while  $\text{Ti}_2\text{C}$  MXene has a clear metallic character. The energy gaps of the isolated clusters disappear upon the composite formation, leading to two completely metallic systems. Note that, upon composite formation, the discrete  $\text{Ti}(3d)$  states from Ti atoms of the titania cluster strongly hybridize by the Mxene  $3d$  band giving rise to a titania band crossing the Fermi level.

Finally, the calculated initial state  $\text{C}(1s)$  and  $\text{O}(1s)$  CLBEs and their  $\Delta\text{CLBEs}$  with respect to graphene  $\text{C}(1s)$  indicate that, upon the titania cluster adsorption, the  $\text{C}(1s)$  of  $\text{Ti}_2\text{C}$  has a small shift of roughly −0.2 eV, while larger shifts are predicted for  $\text{O}(1s)$ . The analysis of the shifts reveal that several factors are likely to contribute to the predicted values which cannot be explained from charge transfer arguments only. Therefore, from a chemical point of view, the information that XPS can provide for these systems is limited. However, it may be useful to experimentalists for characterization purposes.

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## Conflict of Interests

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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